Matthew Ngoy

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Summary

Computer science senior studying trustworthy AI evaluation, verification and sustainable computing.

Education

Kean University Union, NJ
B.S. Computer Science (Cybersecurity Option); GPA: 3.52 *Expected May 2027*

- **Completed Coursework:** Data Structures; Algorithms; Operating Systems; Databases; Computer Security; Cryptography; Ethical Hacking; Digital Forensics.
- **Additional Foundations:** Applied Statistics; Linear Algebra; Environmental Science; Technical Writing. **In Progress:** Fundamentals of AI; Software Engineering; Python; VLM Safety & Red Teaming.
- **Honors & Programs:** NSF IUSE Scholar (2025, 2026); CRA UR2PhD; CAHSI Travel Scholar; Dean's List; Upsilon Pi Epsilon; WELL Program; Young Envoys Scholarship.

Technical Skills

Languages: Python, Java, JavaScript, SQL, PHP, PowerShell
Cybersecurity: OSINT, Nmap, adversarial testing, OWASP LLM Top 10, MITRE ATLAS
Artificial Intelligence & LLM Evaluation: LLM-as-judge, OpenAI API, Ollama
Systems & Development: Linux (Ubuntu, Kali), Git, Docker, MySQL, SQLite, Node.js, Arduino/ESP8266
Coursework Exposure: TypeScript, Wireshark, Metasploit, Hashcat, Autopsy

Experience

Cybersecurity Research Intern Sep. 2026 – Present
Kean University Center for Cybersecurity *Union, NJ*

- Support solar-power integration for Mobile ACDC, a portable teaching data center combining compute, networking and backup power; define whole-rack monitoring requirements.
- Investigate load power, battery capacity and conversion losses to assess solar feasibility; distinguish equipment ratings from measured consumption and identify usable telemetry interfaces.
- Explained phishing, public Wi-Fi risks, password security and digital privacy as a student panelist on WKNJ 90.3 FM's C.Y.B.E.R. Talk.

Cybersecurity Intern (WELL Program) May – Aug. 2026
Kean University Center for Cybersecurity *Union, NJ*

- Classified nonprofit cyber incidents from OSINT, separating confidentiality, integrity and availability impact from CISA sector and TAG Cyber threat categories.
- Integrated browser requests, ESP8266 Wi-Fi communication and Arduino MEGA I/O in a smart-home lab, tracing commands from interface to device control.
- Configured classroom computers for the Port Authority of NY/NJ STEaM Camp, preparing 30 laptops.
- Taught cryptography, programming, and IP addressing through 12 STEaM Camp lessons for 60 high school students.

Projects

CougarCTF 2027 | *Python, Docker, CTFd, GitHub Actions* Sep. 2026 – Present

- Co-develop cryptographic, forensic and reverse-engineering tasks with two teammates for a 200-challenge board; specify necessary stage dependencies so solving requires the intended reasoning.
- Co-develop forensic recovery and execution-state tasks, including reconstruction from superseded database logs and file bytes absent from directory metadata.
- Review alternative solve paths against contestants' tools, rejecting shortcuts that bypass evidence recovery or mathematical-structure recognition.
- Strengthened CI to fail on incomplete reference-solver execution; corrected a leaked flag and 10 hint/text issues from player-view review.

LLM Red-Team Harness | *Python, SQLite, Ollama* June 2026

- Extended rubric-based debate evaluation into adversarial instruction-following tests, comparing exact-canary checks with model judgments across 28 probes and 2 local models.
- Specified persistent per-probe evidence for comparing targets and defense conditions, retaining responses, verdicts and run metadata in JSONL and SQLite.
- Questioned favorable defense results without a meaningful comparison; later re-analysis distinguished filtered-output safety from the underlying model's instruction adherence.

Research & Presentations

HadAgent Verification Study | *Advisor: Dr. Boyang Li* Sep. 2026 – Present

- Identified a score-unit flaw at the verifier's trust boundary: untrusted scale metadata changed comparisons against a fixed acceptance threshold in a copied HadAgent prototype.
- Directed local repair tests using a synthetic grid; enforcing comparable score units reduced below-threshold acceptances from 70/78 to 0/78 while preserving 6/6 synthetic eligible controls.
- Distinguished authenticated claims from proof of model execution; designed independent scoring, eligible controls, and boundary tests to evaluate what verifier acceptance actually establishes.
- Disclosed score-validation and endpoint-trust gaps to the authors; the first author reported fixes, with independent retesting pending.

Gen Z Housing Demographics | *NSF IUSE Scholars Academy* Apr. – Sep. 2026

- Structured ZIP-to-county aggregation with compound geographic keys, population weights and rent-specific nonmissing denominators for demographic–rent analysis across 3,114 U.S. counties.
- Connected MySQL aggregation, PHP/PDO queries and JavaScript charts to investigate young-adult concentration and rent through state comparisons and county drill-down.
- Aligned the unit of analysis before interpreting Pearson correlations; presented the weak county-level association as sole student author at Great Minds in STEM.

Debate Coach LLM Evaluation | *CRA UR2PhD* Feb. – Apr. 2026

- Helped fine-tune debate coaches with two teammates; compared baseline, prompted and adapted conditions to examine whether challenging users' premises supports reconsideration without alienation.
- Refined conversational tone and language complexity to make counterarguments more accessible; tested bugs and model edge cases.
- Evaluated argument strength separately from factual support through rubric scoring and citation checks; contributed interviews on perceived fairness and presented at the CRA UR2PhD showcase.

Adaptive Learning with LLMs | *NSF IUSE Scholars Academy* 2025

- Formulated model and dataset requirements for multi-turn tutoring, stepwise explanations, debugging feedback and follow-up quizzes.
- Encoded learner context and teacher–student dialogue into role-structured training examples; validated content, conversation structure and token limits across 30,738 synthetic records.
- Held the question fixed while varying learner profiles to examine explanation style; presented qualitative findings as first author at Great Minds in STEM.

Leadership

Secretary Sep. 2026 – Present

Women in CyberSecurity (WiCyS), Kean University Chapter *Union, NJ*

- Coordinated PHD4FREE! and co-presented research/funding pathways with Notre Dame PhD student Demetrius Hernandez to 15 attendees.
- Plan WiCyS CTF outreach to universities, companies and prospective speakers.

Peer Mentor / Code Samurai Sep. 2026 – Present; Jan. – May 2025

Kean University Department of Computer Science and Technology *Union, NJ*

- Mentor students in cybersecurity, Python, SQL and data structures, connecting program behavior to concepts through debugging; volunteer in 2025, paid in 2026.

Young Envoys Scholarship Fellow June 2026

Wenzhou-Kean University / WELL Program *Wenzhou and Beijing, China*

- Represented Kean during an educational and cultural exchange; addressed international delegates at a China Education Association for International Exchange event.

International Youth Delegate / IFWY Fellow Sep. – Oct. 2025

International Forum We, the Youth (IFWY) *New York, NY / Seoul, South Korea*

- Proposed SelectSolar to track solar-panel condition and inform reuse or recycling decisions; received 97 public upvotes.
- Specified stakeholder interviews and a pilot comparing predicted with actual panel performance, making validation part of the proposed path from requirements to deployment.
- Recommended extended producer responsibility and contributed environmental recommendations; co-drafted Article I's regional working section in a four-person team.
- Advanced from the fully funded Cornell Tech dialogue to the Seoul final conference through peer election as 1 of 25 North American delegates.